

AI ENGINEER

Zero code to four shipped products in twelve weeks

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AI Engineer — Build and ship 4 real AI products in 12 weeks
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Model names, prices and library versions referenced in this book were accurate at the time of writing and will change. Appendix B covers moving between providers. The techniques do not change.

Companion code, prompt library and evaluation starter pack are available to readers. See the final page for the link.

How to use this book

This is a twelve-week course in book form. Each chapter is one week, and each week ends with a small set of tasks. The tasks are not optional extras — they are where the learning happens. Reading the book without doing them produces the feeling of progress and none of the substance.

Four of the twelve weeks build something you deploy. Weeks 4, 6, 8 and 11 each end with a working application on the internet — four products in total. By the final chapter you have a portfolio, which is the actual product of this book.

You do not need a technical background. You need a computer, an internet connection, about eight hours a week, and roughly five dollars across the whole course for model usage. Chapter 2 sets your machine up from nothing, and Appendix A is a four-page reference sheet for everything in it — keep that open while you work.

Type the code rather than pasting it. You will notice things your eye would skip, and the errors you produce by typing are the fastest teacher in this book. The companion repository exists so you can compare against a working version when you are stuck — not so you can skip the typing.

If you fall behind, do not restart. Skip the reading and do the tasks — they carry the weight.

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About the author

Month One

From zero to a working AI app



01

What an AI engineer actually does

Week 1 — no code in this chapter. Not one line. Read it with a coffee.

1.1 Why this chapter exists

Most books about becoming an AI engineer start by making you install things.

You open page one, and by page three you are copying commands into a black terminal window that you do not understand, watching text scroll past, hoping nothing breaks. Something breaks. You search the error. The answer is on a forum from 2019 and assumes you already know what a virtual environment is. You close the book.

That is not a you problem. That is a bad teaching problem.

Here is what actually happens when someone learns this well. They first build a *mental map* of the territory. They learn what the pieces are called, how the pieces connect, and what a real job in this field looks like on a Tuesday afternoon. Only then do they touch a keyboard. When the code finally arrives, it lands on something. It has a place to sit.

So this chapter has no code. By the end of it you will be able to explain to a friend what an AI engineer does, how it differs from three job titles that sound almost identical, and what the three projects you are about to build actually are. That is a real week of progress, even though nothing gets installed.

A promise, stated honestly.

This book will not make you a senior AI engineer in three months. Nobody can, and anyone selling that is selling a feeling. What this book does is take you from zero code to four shipped, working AI products that live on the internet and that you can show to a person who might hire you. That is achievable in twelve weeks if you show up. It is also, in this particular field right now, often enough to get the first conversation.

1.2 The one-sentence version

An AI engineer builds working products on top of AI models that somebody else trained.

Read that again, because almost every piece of confusion about this career comes from missing it.

You are not going to build an AI. You are not going to sit in a lab with eight graphics cards teaching a machine to understand language from scratch. That work costs tens of millions of dollars and is done by a few thousand people on Earth.

You are going to do the other thing. The thing that pays well, hires constantly, and did not exist as a job title four years ago: taking these extraordinary models that already exist and turning them into something a normal person can use to solve a real problem.



Fig 1.1 • The farm grows the wheat. The kitchen makes the meal.

Think about a restaurant.

Somewhere far away, there are farms. Enormous, industrial, capital-intensive farms that grow wheat and raise chickens. That is the AI research lab — the handful of organisations spending fortunes to produce the raw ingredient.

Then there is the restaurant kitchen. The chef did not grow the wheat. The chef takes the ingredient, understands deeply what it can and cannot do, combines it with other things, controls the heat, plates it, and hands a customer something they actually want to eat.

You are the chef. The model is flour.

And notice: nobody thinks the chef is doing lesser work than the farmer. They are different work. The chef's skill is *judgment* — knowing what to make, what

combines with what, when it is undercooked, and what the person sitting at the table actually wanted.

1.3 Four job titles that sound the same

This trips up every beginner, and it will come up in your first interview. Learn it now.



Fig 1.2 · Four desks. Only one of them is yours.

The AI Researcher

Invents new kinds of models. Usually holds a PhD. Publishes papers. Works at a lab or a university. Asks: *What if we structured the model this way instead?*

The Data Scientist

Finds patterns in a company’s existing data to inform decisions. Lives in spreadsheets, statistics and charts. Asks: *Why did customer churn increase last quarter?*

The ML Engineer

Trains and fine-tunes models on specific data, and keeps them running at scale. Sits closest to traditional software engineering plus statistics. Asks: *How do we train this on our own data and serve it to a million users?*

The AI Engineer — you

Builds applications using existing models through their APIs. Asks: *How do I turn this model into a product a person will pay for?*

The reason your role is hiring so aggressively is simple arithmetic. There are a handful of labs producing models. There are millions of companies who need software built with them. The bottleneck is not model creation. The bottleneck is people who can build the thing on top.

	PhD?	Trains models?	Product code?	Entry path
AI Researcher	Almost always	Yes, invents them	Rarely	Academia
Data Scientist	Sometimes	Sometimes	Some	Stats / analytics
ML Engineer	No	Yes	Yes	Software engineering
AI Engineer	No	No	Yes	Portfolio of built things

That last row is the whole reason this book can exist. Your entry path is not a credential. It is evidence. You get in by having built things, which means you can start today.

1.4 What the job looks like on a Tuesday

Abstract descriptions are useless. Here is a real day.

A logistics company has a problem. Their support team gets four hundred emails a day, and roughly seventy percent are the same eight questions: where is my shipment, how do I change a delivery address, what are your rates to Singapore. Agents spend their day copy-pasting.

You are hired to fix it. Your Tuesday:

9:30am. You read forty real support emails. Not code — reading. You are figuring out what the eight questions actually are and where the answers live. Half of them are in a PDF nobody has updated since 2023. This is the job.

11:00am. You write instructions for the model. Not a program — instructions, in English. You test it. It confidently invents a shipping rate. You tighten the instructions. This is prompt engineering, and you will get good at it in Week 3.

1:30pm. The model needs access to the company's actual documents, because it has never seen them. You set up a system that finds the three most relevant paragraphs and hands them to the model before it answers. This is called RAG, it is the single most valuable thing in this book, and you build one in Week 6.

3:00pm. For *where is my shipment*, no document helps — the answer lives in a live database. So you give the model the ability to look it up itself. You describe a tool; the model decides when to call it. Tool calling, Week 7.

4:30pm. You check the bill. Every request costs a fraction of a rupee, but four hundred a day adds up, and you notice one badly written instruction is tripling the cost of every call. You fix it. Nobody teaches this, and it is why some engineers get promoted. Week 9.

5:15pm. You run fifty test emails and find the model still guesses at rates. You add a hard rule. You run them again.

That is the job. Notice how much of it is reading, judgment and testing — and how little is typing clever code.

1.5 The three things you will build



Fig 1.3 · Twelve weeks, three plateaus, one flagship.

By Week 12 you will have three things on the internet with your name on them.

Project 1, Week 4 — a tool that does one job well

Small, complete, real. You pick the job. Turning messy meeting notes into clean action items is a good one. It will take you an afternoon by Week 4 and it will feel like magic, because six weeks earlier you had never written a line of code.

Project 2, Week 6 — chat with your own documents

You upload files — a textbook, a contract, a company handbook — and ask questions in plain language. The answers come from *your* documents, not from the model's imagination. This is the highest-demand pattern in the industry right now, and freelancers charge real money for exactly this.

Project 3, Week 8 — an agent that does a job

Not a chatbot that talks. Something that acts: reads incoming messages, decides what to do, uses tools, and reports back. This is where people lean forward when you demo it.

Flagship, Week 11 — a full product

Your three pieces, combined, deployed, with a proper interface, secured, and live at a URL you can put on a résumé.

1.6 The honest part

Three things that will be true, so they do not surprise you later.

Week 2 is the hard one

It is the Python week, and it is where non-technical readers quit. So here is the deal I am making with you: I will never ask you to invent code from a blank page. Every line you type will be explained, and you will always be modifying something that already works. Programming is far more like editing than like writing.

The model will lie to you, confidently

It will state a wrong fact in a beautiful, well-structured sentence. This is called hallucination, it is not a bug that will be fixed next year, and half of professional AI engineering is building systems that constrain it. Once you internalise that, you stop being fooled and start being useful.